Advanced Programming Practice · 21CSC203P

Unit 2 · Session 10 · SLO 1 — Solved Worksheet

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Unit – 2 : Session – 10 : SLO – 1 | Inheritance with Overriding

Activity (Solution)

• class Figure { double dim1, dim2; Figure(double a, double b){ dim1=a; dim2=b; } double area(){ System.out.println("Area undefined for Figure"); return 0; } }

• class Rectangle extends Figure { Rectangle(double l, double w){ super(l, w); } @Override double area(){ return dim1 * dim2; } }

• class Triangle extends Figure { Triangle(double b, double h){ super(b, h); } @Override double area(){ return 0.5 * dim1 * dim2; } }

• public class Main { public static void main(String[] a){ Figure f; f = new Rectangle(10, 5); System.out.println("Rectangle area: " + f.area()); f = new Triangle(10, 5); System.out.println("Triangle area: " + f.area()); } }

• Output: Rectangle area: 50.0 / Triangle area: 25.0